PONTIFÍCIA UNIVERSIDADE CATÓLICA DO RIO GRANDE DO SUL
ESCOLA POLITÉCNICA

MALTA
Machine Learning Theory
and Applications Lab

Aprendizado Profundo II

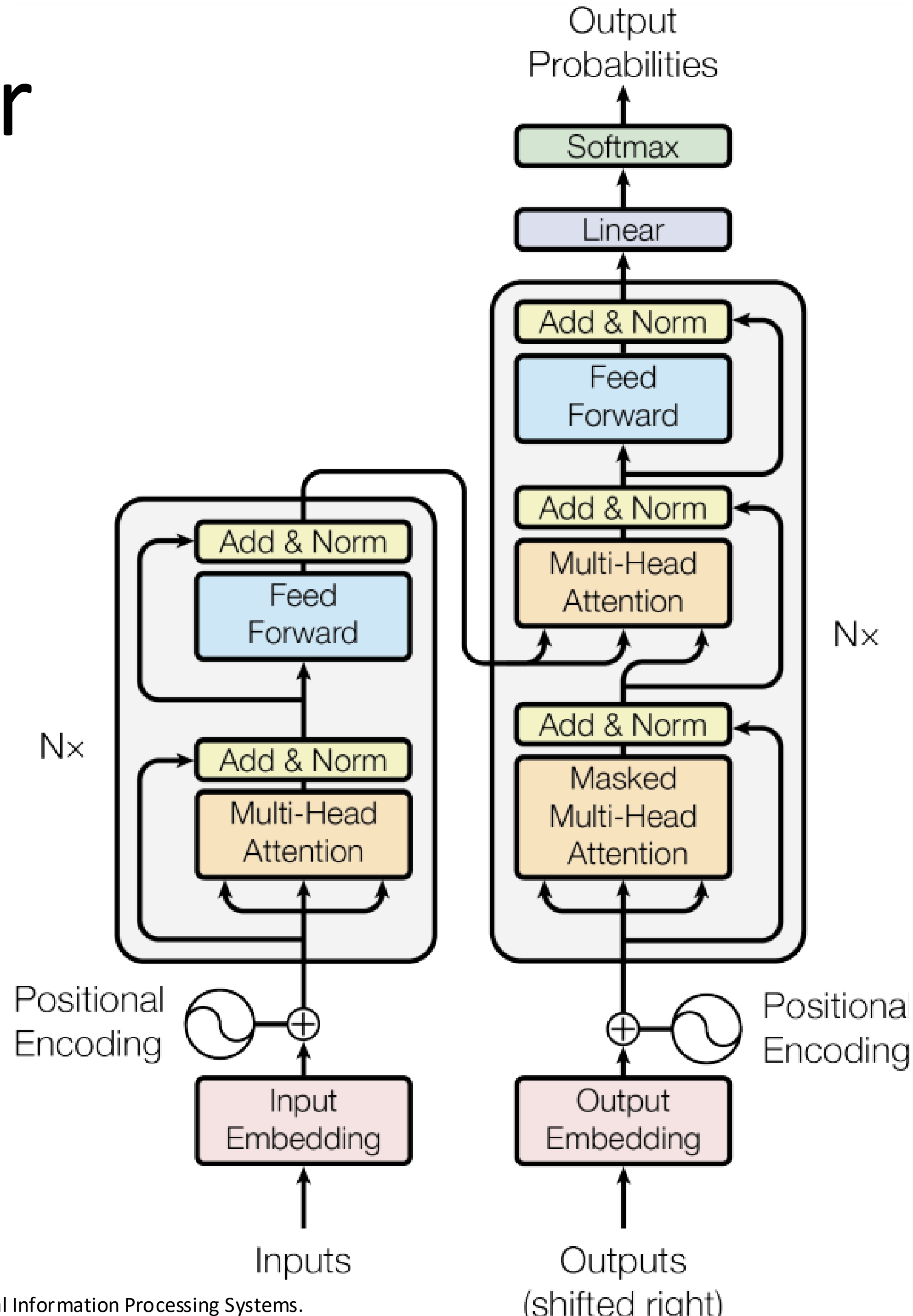
GPT, Llama e Alinhamento

Prof. Me. Otávio Parraga

Roteiro

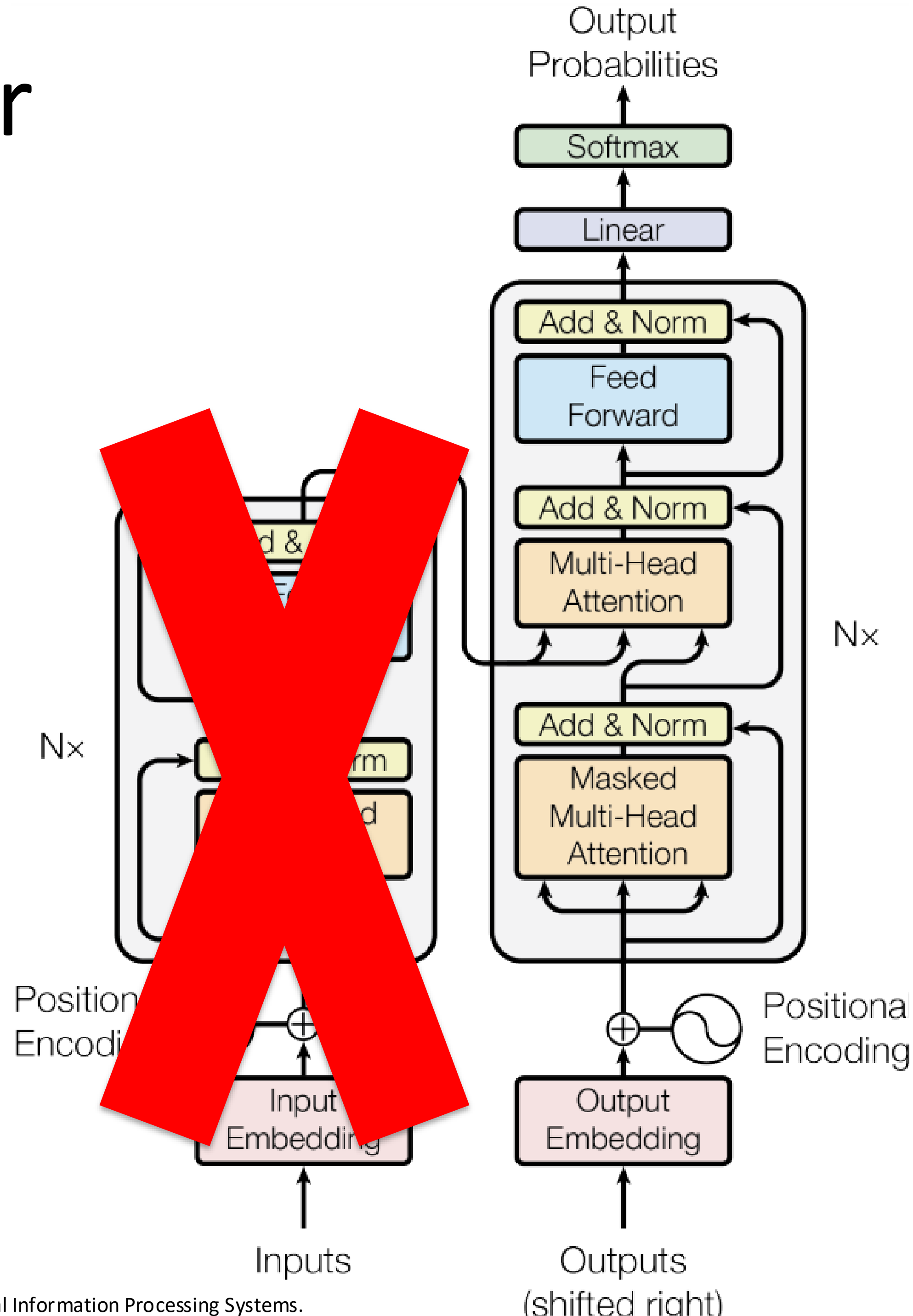
- Família GPT
- LLMs
- Alinhamento

Transformer Decoder



Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

Transformer Decoder



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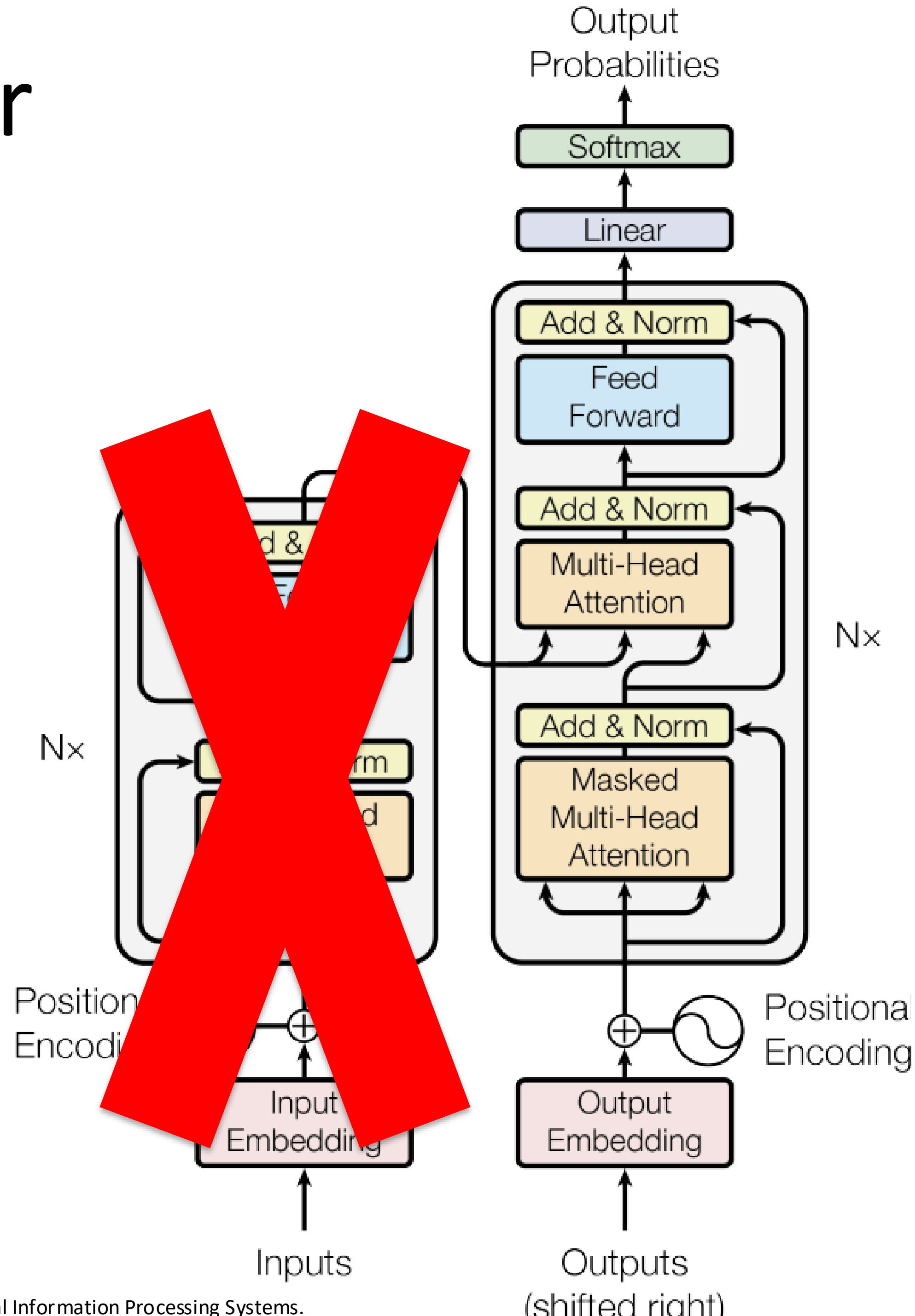
Transformer Decoder

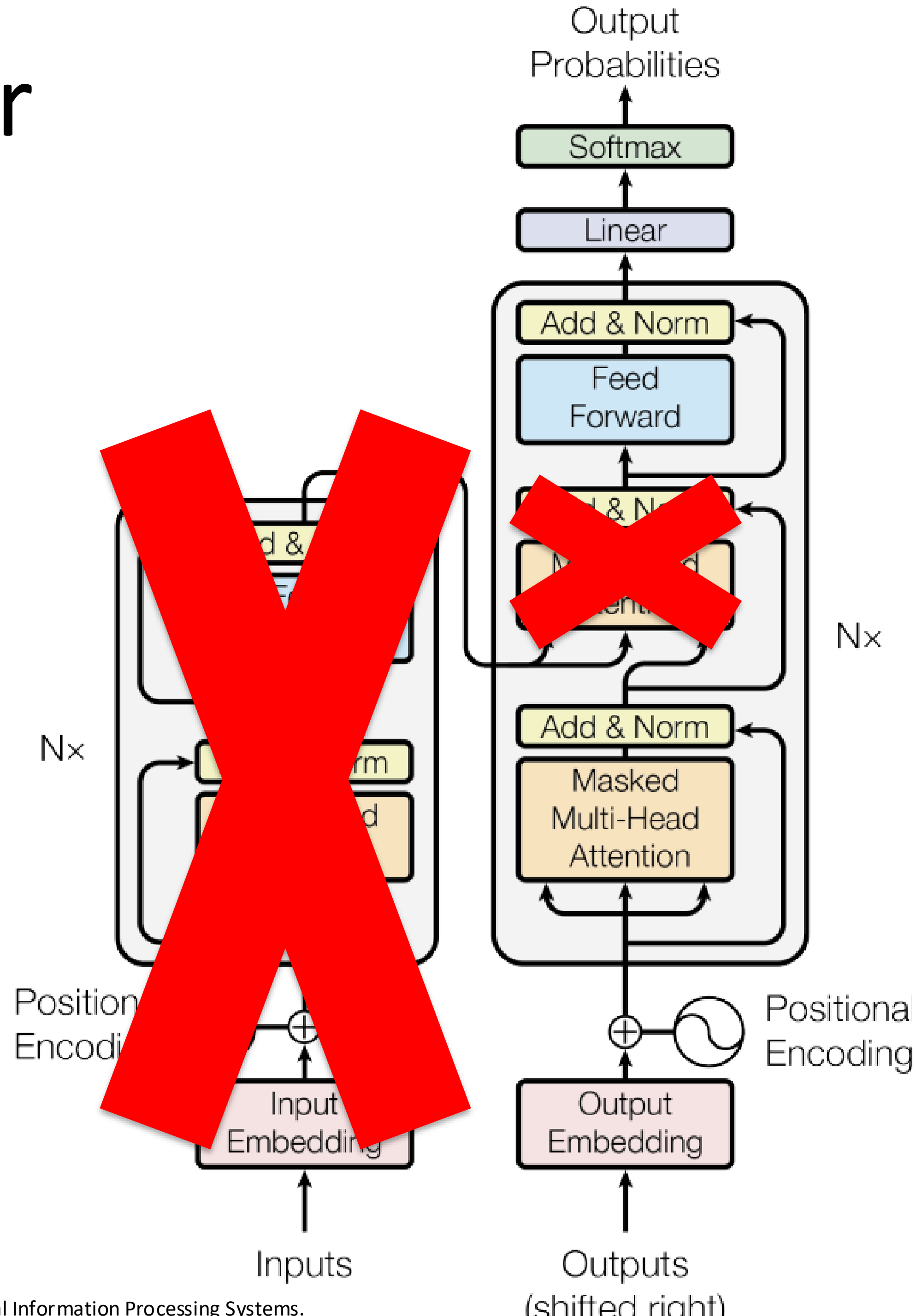


Problema:
Temos uma sublayer
que precisa do encoder

Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

Transformer Decoder

Problema:
Temos uma sublayer
que precisa do encoder
Solução:
Removemos ela
também



Vaswani, A. (2017). Attention is all you need. Advances in Neural Information Processing Systems.

Família GPT

- Generative Pre-Training
- Lançado em 2018 pela equipe da OpenAI (Quando ainda era "Open")
- Termo GPT nunca usado no paper original

Improving Language Understanding by Generative Pre-Training

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Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

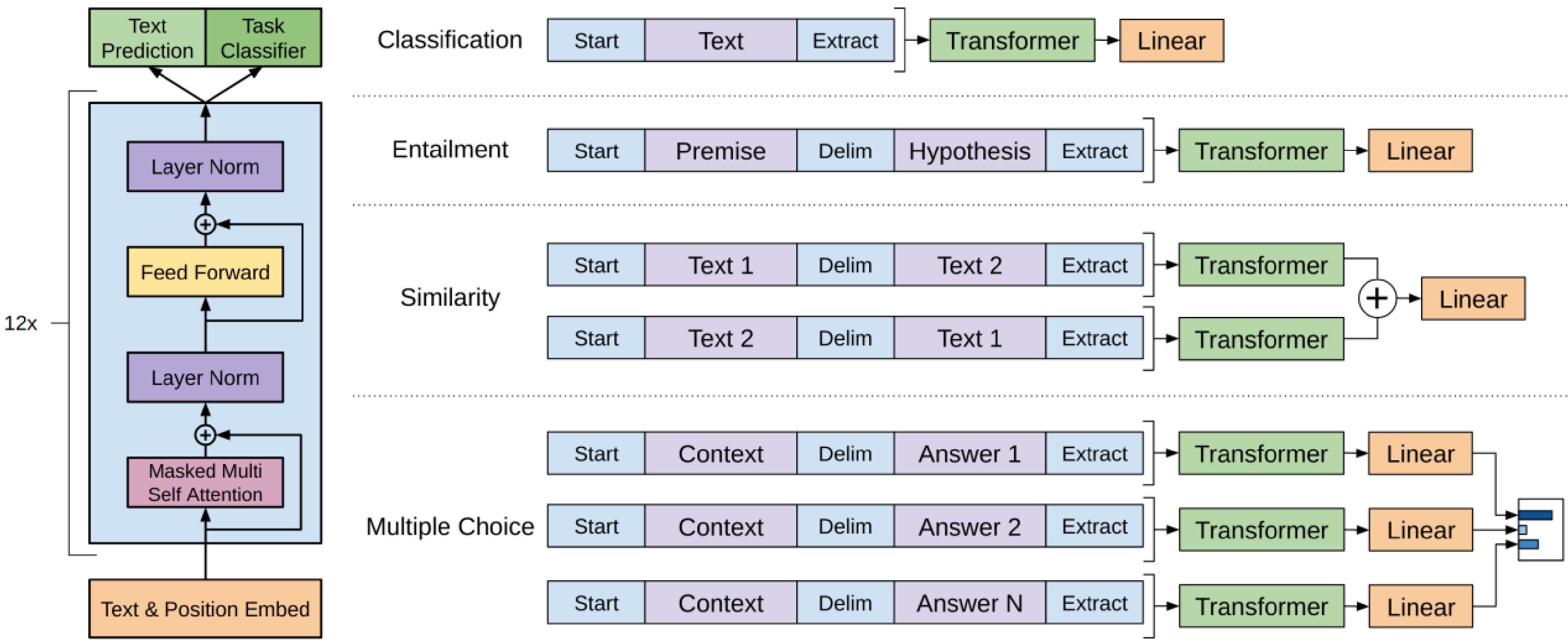
Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.[1] Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

[1]https://gluebenchmark.com/leaderboard

Família GPT

- Trouxe a mesma ideia do BERT -> Pré-treino e posteriormente *Fine-tuning*
- Pré-treinado com Continuous Language Modeling
  - Comumente referido como modelo Autorregressivo
  - *Cross-Entropy*
- *Fine-tuning* com a tarefa alvo e também com CLM

GPT-1



Radford, Alec, et al. "Improving language understanding by generative pre-training." (2018).

GPT-1

- Treinamento em cima do BookCorpus
- Inúmeros datasets abordados na avaliação, lidando com: NLI, Q&A, Sentence Similarity, Classification
- 512 tokens de entrada
- Função de Ativação GELU
- Cosine Scheduler
- Positional Embeddings Aprendidos
- 12 camadas de Decoder
- 117M de parâmetros

Radford, Alec, et al. "Improving language understanding by generative pre-training." (2018).

GPT-2

- *"Language Models are Unsupervised Miltitask Learners"* (2019)
- Exato mesmo modelo do GPT-1
- Mais dados de treinamento e mais parâmetros
  - WebText – 8M de documentos

Language Models are Unsupervised Multitask Learners

Alec Radford [*1] Jeffrey Wu [*1] Rewon Child [1] David Luan [1] Dario Amodei [**1] Ilya Sutskever [**1]

Abstract

Natural language processing tasks, such as question answering, machine translation, reading comprehension, and summarization, are typically approached with supervised learning on task-specific datasets. We demonstrate that language models begin to learn these tasks without any explicit supervision when trained on a new dataset of millions of webpages called WebText. When conditioned on a document plus questions, the answers generated by the language model reach 55 F1 on the CoQA dataset - matching or exceeding the performance of 3 out of 4 baseline systems without using the 127,000+ training examples. The capacity of the language model is essential to the success of zero-shot task transfer and increasing it improves performance in a log-linear fashion across tasks. Our largest model, GPT-2, is a 1.5B parameter Transformer that achieves state of the art results on 7 out of 8 tested language modeling datasets in a zero-shot setting but still underfits WebText. Samples from the model reflect these improvements and contain coherent paragraphs of text. These findings suggest a promising path towards building language processing systems which learn to perform tasks from their naturally occurring demonstrations.

1. Introduction

Machine learning systems now excel (in expectation) at tasks they are trained for by using a combination of large datasets, high-capacity models, and supervised learning (Krizhevsky et al., 2012) (Sutskever et al., 2014) (Amodei et al., 2016). Yet these systems are brittle and sensitive to slight changes in the data distribution (Recht et al., 2018) and task specification (Kirkpatrick et al., 2017). Current systems are better characterized as narrow experts rather than competent generalists. We would like to move towards more general systems which can perform many tasks – eventually without the need to manually create and label a training dataset for each one.

The dominant approach to creating ML systems is to collect a dataset of training examples demonstrating correct behavior for a desired task, train a system to imitate these behaviors, and then test its performance on independent and identically distributed (IID) held-out examples. This has served well to make progress on narrow experts. But the often erratic behavior of captioning models (Lake et al., 2017), reading comprehension systems (Jia & Liang, 2017), and image classifiers (Alcorn et al., 2018) on the diversity and variety of possible inputs highlights some of the shortcomings of this approach.

Our suspicion is that the prevalence of single task training on single domain datasets is a major contributor to the lack of generalization observed in current systems. Progress towards robust systems with current architectures is likely to require training and measuring performance on a wide range of domains and tasks. Recently, several benchmarks have been proposed such as GLUE (Wang et al., 2018) and decaNLP (McCann et al., 2018) to begin studying this.

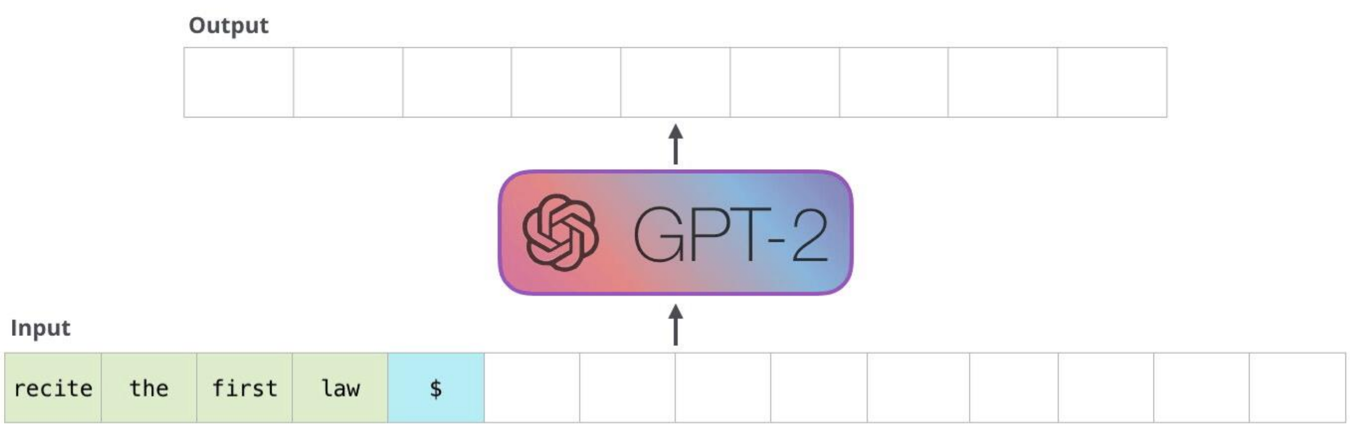
Multitask learning (Caruana, 1997) is a promising framework for improving general performance. However, multitask training in NLP is still nascent. Recent work reports modest performance improvements (Yogatama et al., 2019) and the two most ambitious efforts to date have trained on a total of 10 and 17 `(dataset, objective)` pairs respectively (McCann et al., 2018) (Bowman et al., 2018). From a meta-learning perspective, each `(dataset, objective)` pair is a single training example sampled from the distribution of datasets and objectives. Current ML systems need hundreds to thousands of examples to induce functions which generalize well. This suggests that multitask training many need just as many effective training pairs to realize its promise with current approaches. It will be very difficult to continue to scale the creation of datasets and the design of objectives to the degree that may be required to brute force our way there with current techniques. This motivates exploring additional setups for performing multitask learning.

*,**Equal contribution [1]OpenAI, San Francisco, California, United States. Correspondence to: Alec Radford <alec@openai.com>.

Radford, Alec, et al. "Language models are unsupervised multitask learners." *OpenAI blog* 1.8 (2019): 9.

GPT-2

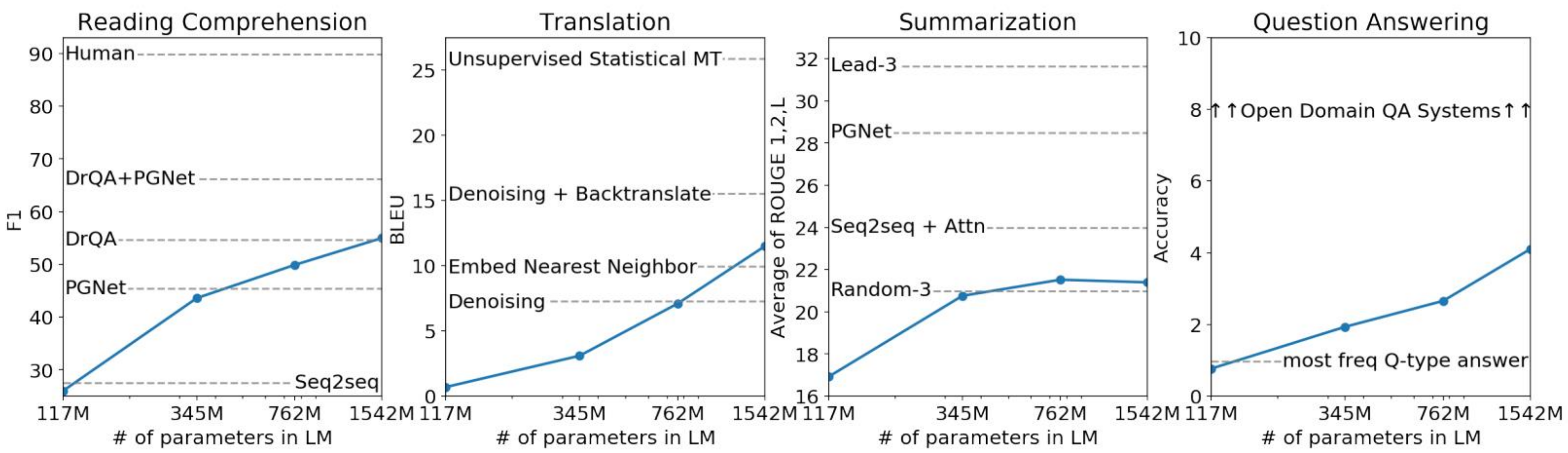
- Maior versão com 1.5B de parâmetros
  - 48 camadas
- Entrada de 1024 tokens
- Primeiro modelo transformer a gerar texto coerente semântica e sintaticamente



Radford, Alec, et al. "Language models are unsupervised multitask learners." *OpenAI blog* 1.8 (2019): 9.
https://jalammar.github.io/illustrated-gpt2/

GPT-2

- Por que Multitask Learners?
  - Autores defendem que não é preciso o *fine-tuning*
  - Tarefas são aprendidas apenas com *Language Modeling*
  - Resultados do paper relatados sem nenhum *fine-tuning*
    - *Sempre fornecendo o documento e condicionando o modelo para a tarefa desejada*



Radford, Alec, et al. "Language models are unsupervised multitask learners." *OpenAI blog* 1.8 (2019): 9.

GPT-3

- Aumentamos ainda mais os tamanhos de modelo, dataset e tempo de processamento
  - 175B maior modelo
  - 45TB de dados de treinamento

arXiv:2005.14165v4 [cs.CL] 22 Jul 2020

Language Models are Few-Shot Learners

Tom B. Brown* Benjamin Mann* Nick Ryder* Melanie Subbiah*

Jared Kaplan† Prafulla Dhariwal Arvind Neelakantan Pranav Shyam Girish Sastry

Amanda Askell Sandhini Agarwal Ariel Herbert-Voss Gretchen Krueger Tom Henighan

Rewon Child Aditya Ramesh Daniel M. Ziegler Jeffrey Wu Clemens Winter

Christopher Hesse Mark Chen Eric Sigler Mateusz Litwin Scott Gray

Benjamin Chess Jack Clark Christopher Berner

Sam McCandlish Alec Radford Ilya Sutskever Dario Amodei

OpenAI

Abstract

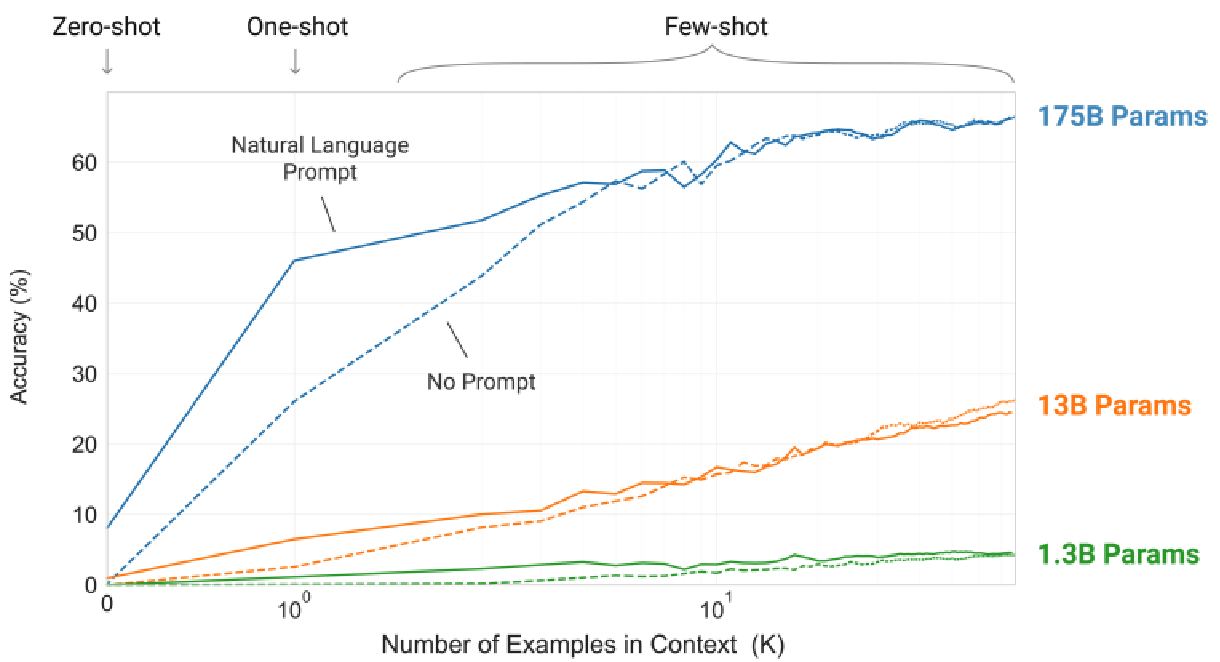
Recent work has demonstrated substantial gains on many NLP tasks and benchmarks by pre-training on a large corpus of text followed by fine-tuning on a specific task. While typically task-agnostic in architecture, this method still requires task-specific fine-tuning datasets of thousands or tens of thousands of examples. By contrast, humans can generally perform a new language task from only a few examples or from simple instructions – something which current NLP systems still largely struggle to do. Here we show that scaling up language models greatly improves task-agnostic, few-shot performance, sometimes even reaching competitiveness with prior state-of-the-art fine-tuning approaches. Specifically, we train GPT-3, an autoregressive language model with 175 billion parameters, 10x more than any previous non-sparse language model, and test its performance in the few-shot setting. For all tasks, GPT-3 is applied without any gradient updates or fine-tuning, with tasks and few-shot demonstrations specified purely via text interaction with the model. GPT-3 achieves strong performance on many NLP datasets, including translation, question-answering, and cloze tasks, as well as several tasks that require on-the-fly reasoning or domain adaptation, such as unscrambling words, using a novel word in a sentence, or performing 3-digit arithmetic. At the same time, we also identify some datasets where GPT-3's few-shot learning still struggles, as well as some datasets where GPT-3 faces methodological issues related to training on large web corpora. Finally, we find that GPT-3 can generate samples of news articles which human evaluators have difficulty distinguishing from articles written by humans. We discuss broader societal impacts of this finding and of GPT-3 in general.

*Equal contribution
†Johns Hopkins University, OpenAI

Brown, Tom, et al. "Language models are few-shot learners." *Advances in neural information processing systems* 33 (2020): 1877-1901.

Large Language Models (LLMs)

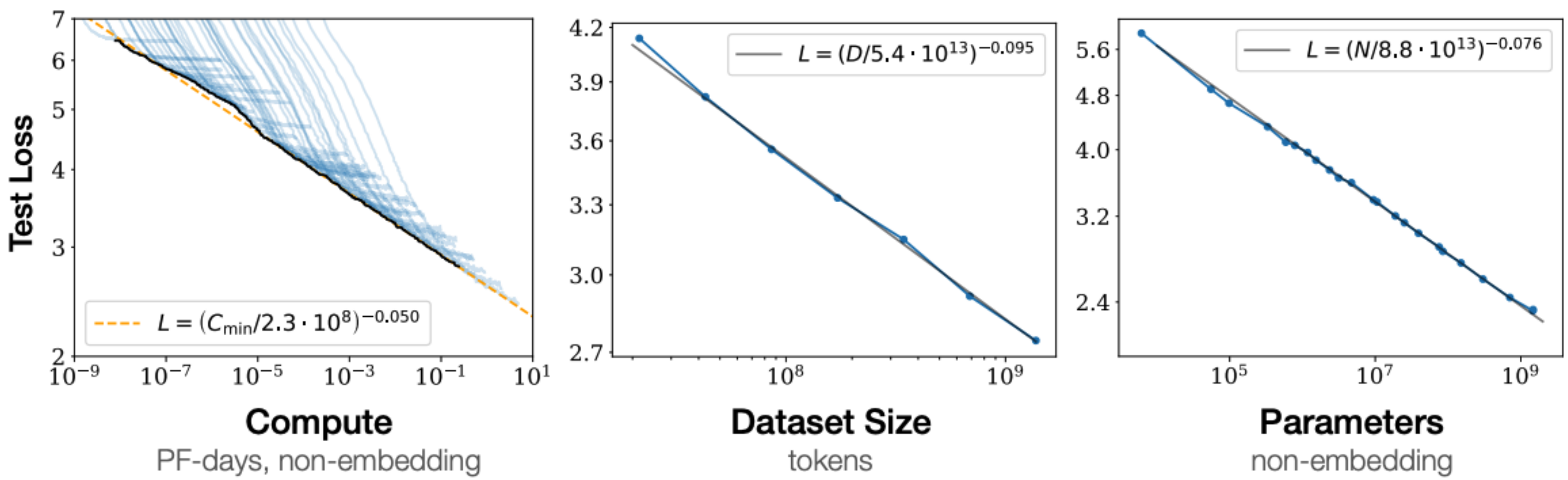
- Na prática, GPT-3 foi o primeiro paper a trazer o conceito de um Large Language Model (LLM)
- Temos um modelo capaz de executar inúmeras tarefas sem ter sido explicitamente treinado para elas



Brown, Tom, et al. "Language models are few-shot learners." *Advances in neural information processing systems* 33 (2020): 1877-1901.

Large Language Models (LLMs)

- GPT-3 trouxe o que veio a ser chamado de capacidades emergentes
  - Possibilidade de fazer Few-shot e Zero-shot learning
- Percebemos que Transformers seguem regras específicas de escalonamento (*Scaling Laws*)



Kaplan, J., McCandlish, S., Henighan, T., Brown, T. B., Chess, B., Child, R., ... & Amodei, D. (2020). Scaling laws for neural language models. arXiv preprint arXiv:2001.08361.

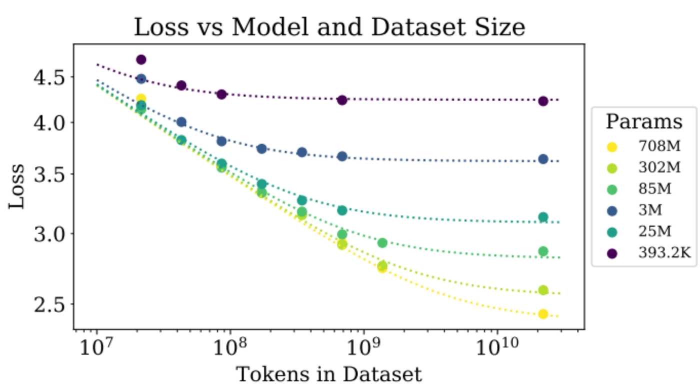
Large Language Models (LLMs)

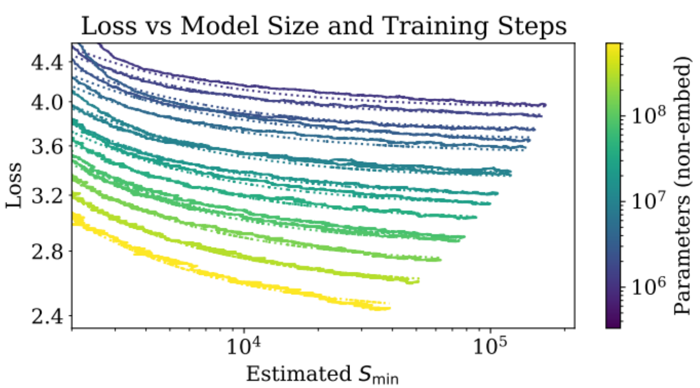
- Capacidades emergentes são aquelas não esperadas em modelos menores
- "*Emergence is when quantitative changes in a system result in qualitative changes in behavior*" – Phillip Anderson

Wei, Jason, et al. "Emergent abilities of large language models." *arXiv preprint arXiv:2206.07682* (2022).

Large Language Models (LLMs)

- Conforme aumentamos a quantidade de dados de treinamento, computação e parâmetros, modelos de linguagem conseguem valores menores de loss
  - Geralmente, aumentos sutis de performance
- Principal fator é a escala, pouco sensível a decisões arquiteturais





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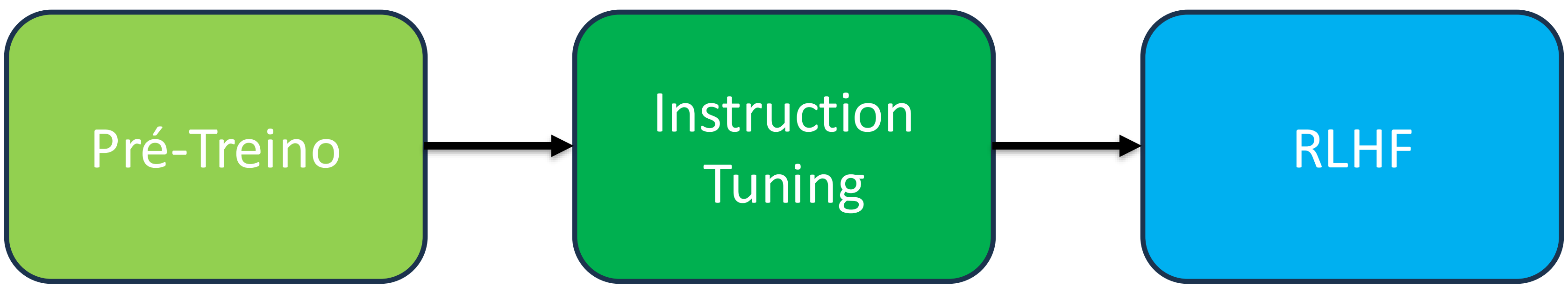
Large Language Models (LLMs)

- Porém, treinar para LM não torna os modelos bons agentes conversacionais...
- Para termos um modelo como o ChatGPT precisamos de algumas etapas adicionais:



Large Language Models (LLMs)

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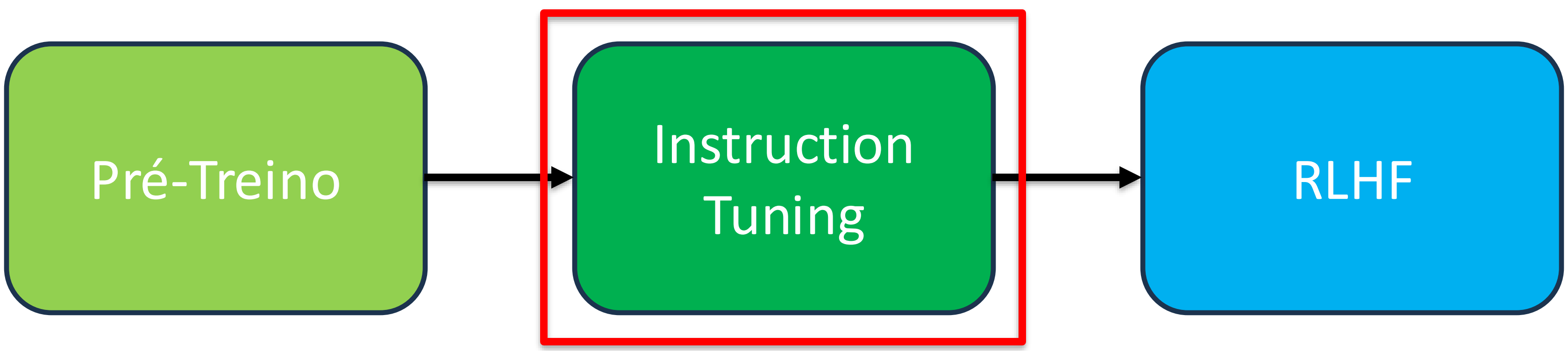


Instruction Tuning

- Treinamos para LM, logo ...

PROMPT	*Explain the moon landing to a 6 year old in a few sentences.*
COMPLETION	GPT-3
	`Explain the theory of gravity to a 6 year old.`
	`Explain the theory of relativity to a 6 year old in a few sentences.`
	`Explain the big bang theory to a 6 year old.`
	`Explain evolution to a 6 year old.`

Ouyang, Long, et al. "Training language models to follow instructions with human feedback." *Advances in neural information processing systems* 35 (2022): 27730-27744.

Instruction Tuning

- Treinamos para LM, logo ...
- Mas como usuários, esperamos algo diferente...

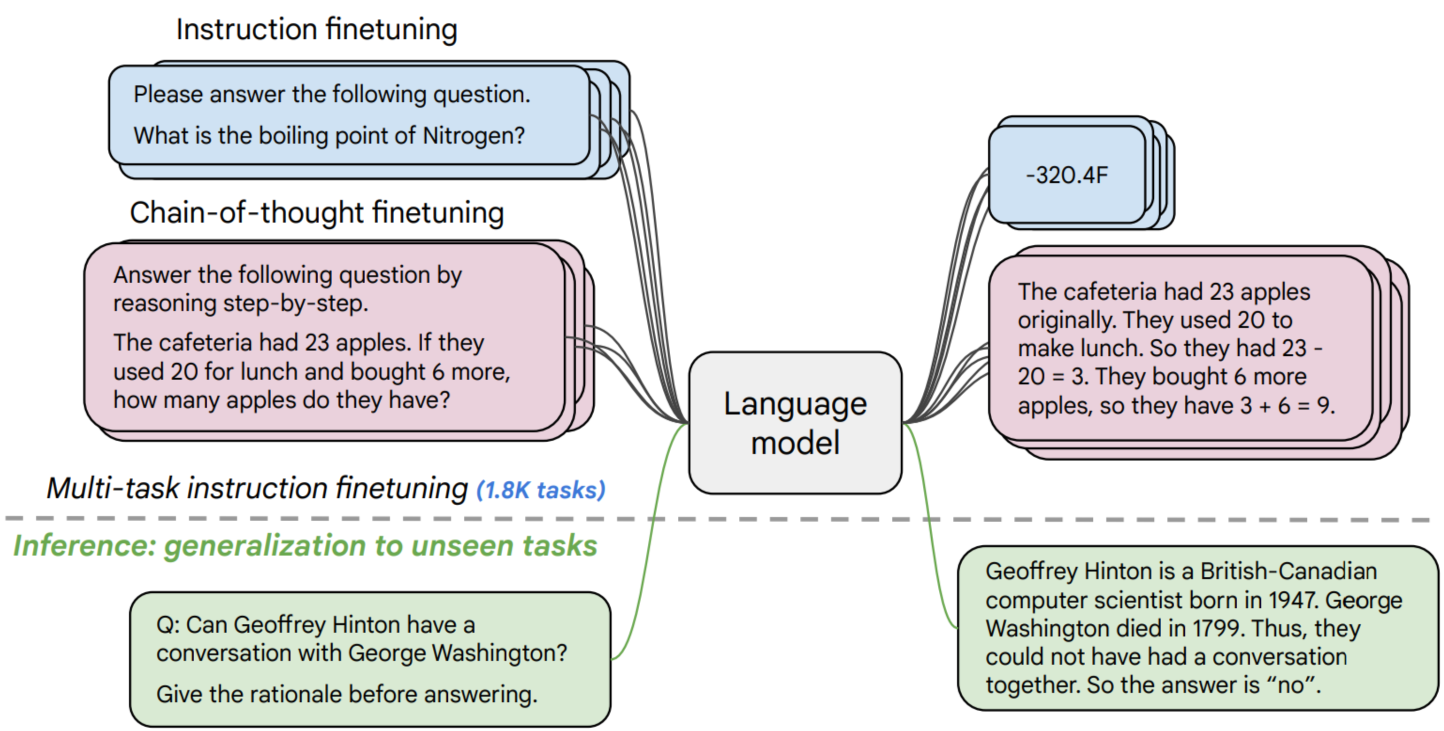
PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION **Human**
A giant rocket ship blasted off from Earth carrying astronauts to the moon. The astronauts landed their spaceship on the moon and walked around exploring the lunar surface. Then they returned safely back to Earth, bringing home moon rocks to show everyone.

Ouyang, Long, et al. "Training language models to follow instructions with human feedback." *Advances in neural information processing systems* 35 (2022): 27730-27744.

Instruction Tuning

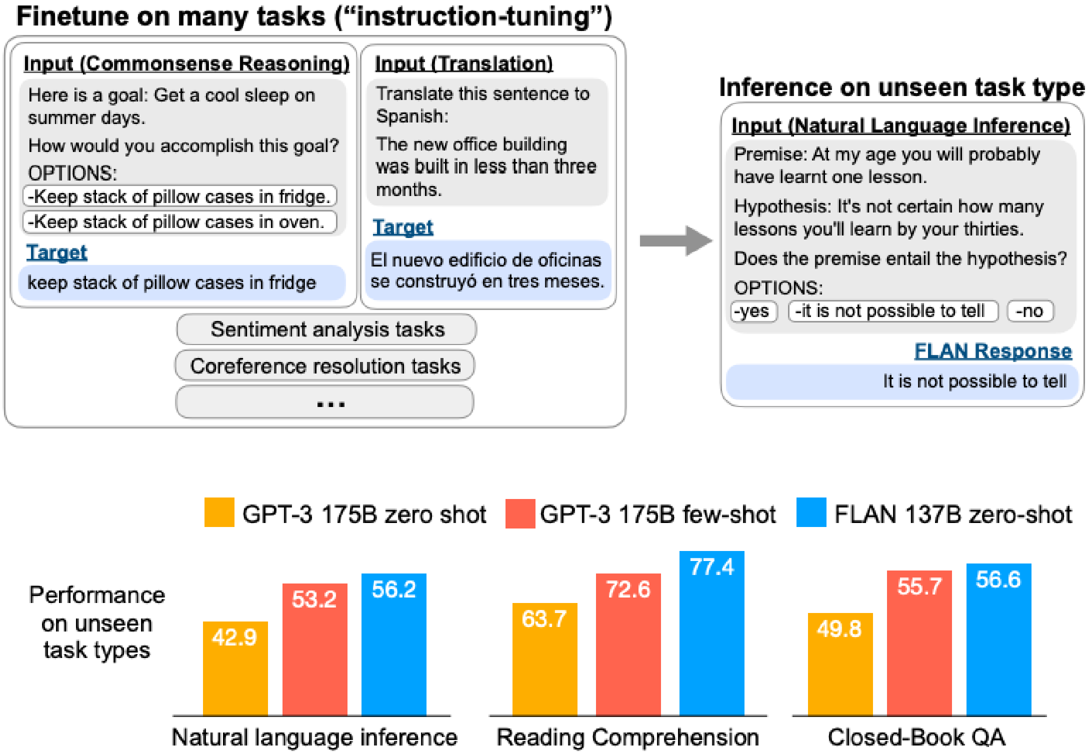
- Como resolvemos o problema?
  - Podemos fazer um *fine-tuning* específico:



Chung, Hyung Won, et al. "Scaling instruction-finetuned language models." *Journal of Machine Learning Research* 25.70 (2024): 1-53.

Instruction Tuning

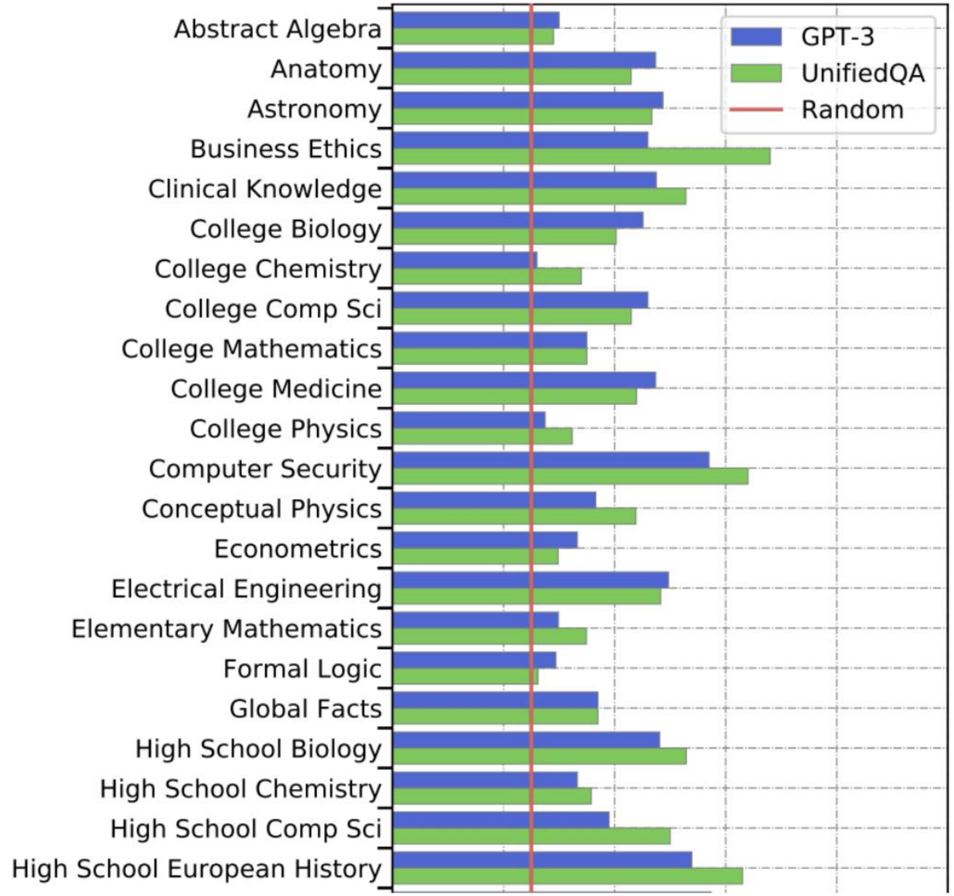
- Ensinamos o modelo a responder perguntas de diferentes tarefas
  – Treinamento feito com *Supervised Fine-Tuning*



Wei, Jason, et al. "Finetuned language models are zero-shot learners." *arXiv preprint arXiv:2109.01652* (2021).

Instruction Tuning

- Para avaliar, usamos tarefas não vistas
- Incluímos grandes Benchmarks como Massive Multitask Language Understanding (MMLU)



Hendrycks, Dan, et al. "Measuring massive multitask language understanding." *arXiv preprint arXiv:2009.03300* (2020).

Instruction Tuning

- Exemplo:

Astronomy

What is true for a type-Ia supernova?

A. This type occurs in binary systems.
B. This type occurs in young galaxies.
C. This type produces gamma-ray bursts.
D. This type produces high amounts of X-rays.
Answer: A

High School Biology

In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

A. directional selection.
B. stabilizing selection.
C. sexual selection.
D. disruptive selection
Answer: A

Hendrycks, Dan, et al. "Measuring massive multitask language understanding." *arXiv preprint arXiv:2009.03300* (2020).

Instruction Tuning

- Resultado de Instruction Tuning:

Model input (Disambiguation QA)

Q: In the following sentences, explain the antecedent of the pronoun (which thing the pronoun refers to), or state that it is ambiguous.

Sentence: The reporter and the chef will discuss their favorite dishes.

Options:
(A) They will discuss the reporter's favorite dishes
(B) They will discuss the chef's favorite dishes
(C) Ambiguous

A: Let's think step by step.

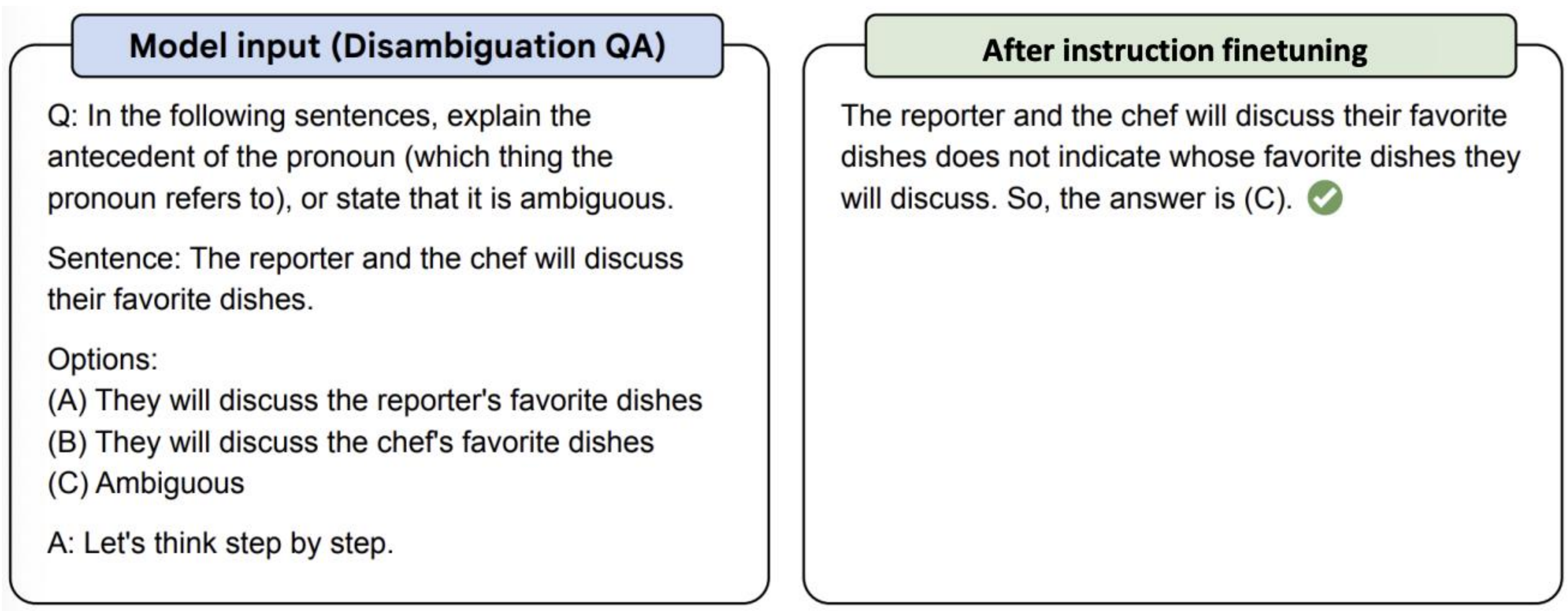
Before instruction finetuning

The reporter and the chef will discuss their favorite dishes.
The reporter and the chef will discuss the reporter's favorite dishes.
The reporter and the chef will discuss the chef's favorite dishes.
The reporter and the chef will discuss the reporter's and the chef's favorite dishes.

✖ **(doesn't answer question)**

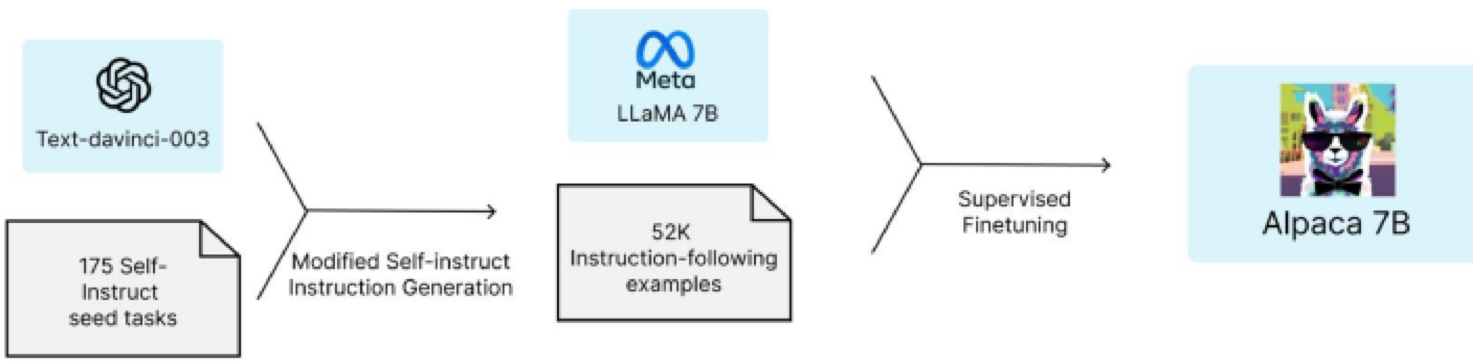
Instruction Tuning

- Resultado de Instruction Tuning:



Instruction Tuning

- Processo de Instrcution Tuning pode usar dados sintéticos
  - Modelo e conjunto de dados ALPACA



Wang, Y., Kordi, Y., Mishra, S., Liu, A., Smith, N. A., Khashabi, D., & Hajishirzi, H. (2022). Self-instruct: Aligning language models with self-generated instructions. *arXiv preprint arXiv:2212.10560*.

Instruction Tuning

- Na prática, também não precisamos de muitos exemplos para fazer o instruction-tuning

Source	#Examples
Training	
Stack Exchange (STEM)	200
Stack Exchange (Other)	200
wikiHow	200
Pushshift r/WritingPrompts	150
Natural Instructions	50
Paper Authors (Group A)	200

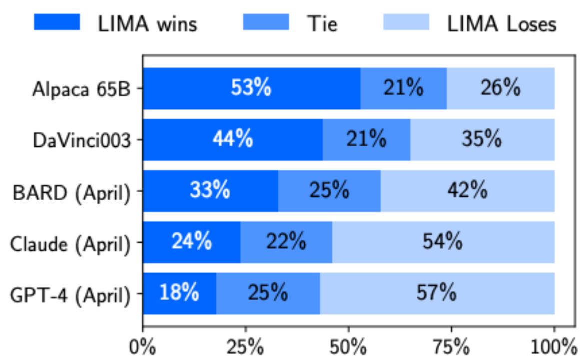


Figure 1: Human preference evaluation, comparing LIMA to 5 different baselines across 300 test prompts.

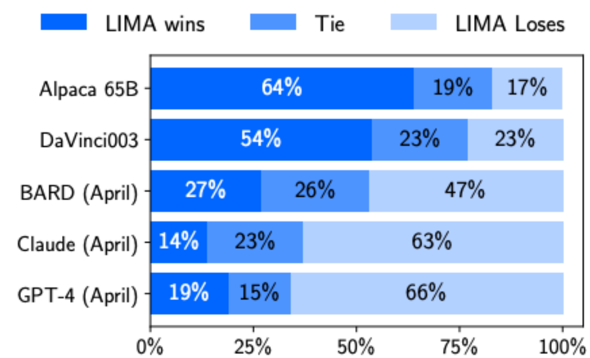


Figure 2: Preference evaluation using GPT-4 as the annotator, given the same instructions provided to humans.

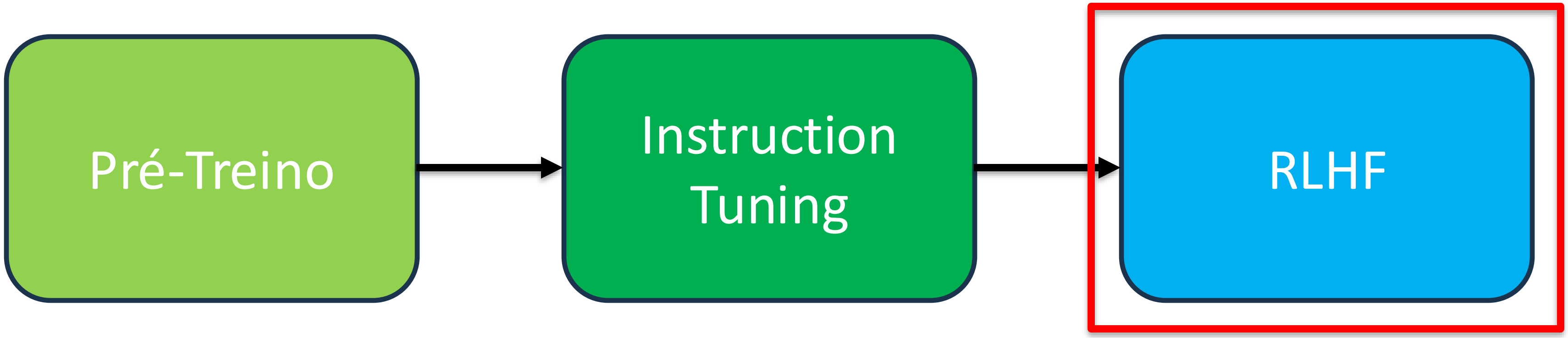
Zhou, Chunting, et al. "Lima: Less is more for alignment." *Advances in Neural Information Processing Systems* 36 (2024).

Instruction Tuning

- Mas nem tudo é resolvido com Instruction-Tuning
  - Tarefas que não possuam uma resposta certa (criatividade) não podem ser melhoradas
  - Alguns erros são piores que outros, e nessa etapa, consideramos todos de forma igual

Large Language Models (LLMs)

- Porém, treinar para LM não torna os modelos bons agentes conversacionais...
- Para termos um modelo como o ChatGPT precisamos de algumas etapas adicionais:

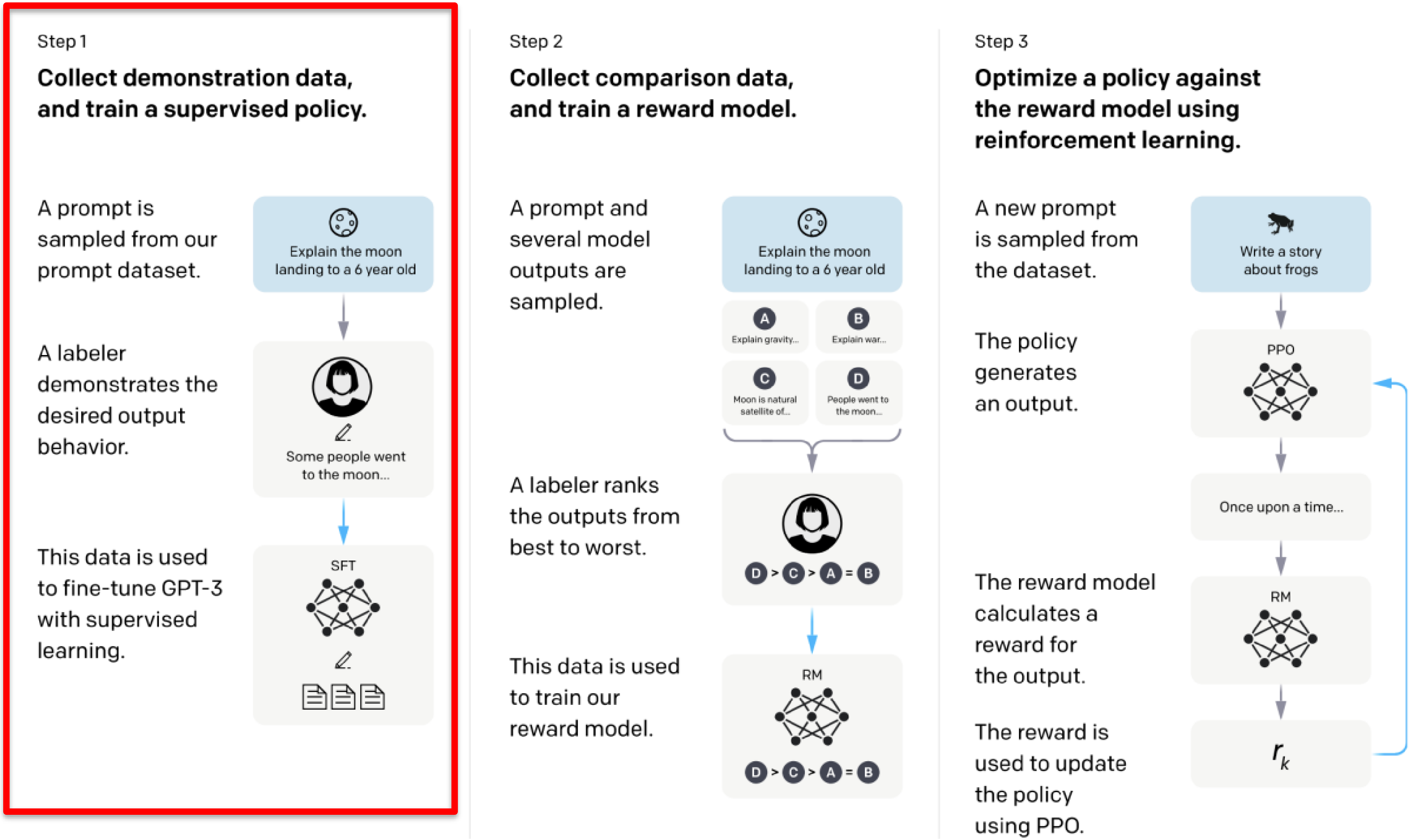


Reinforcement Learning by Human Feedback

- A ideia:
  - Vamos supor que queremos treinar um modelo bom em sumarização
  - E que para cada prompt s vamos querer uma "recompensa humana" para avaliar o quão bom foi o resumo: $R(s) \in \mathbb{R}$
  - Vamos usar essa recompensa para refinar nosso LLM!

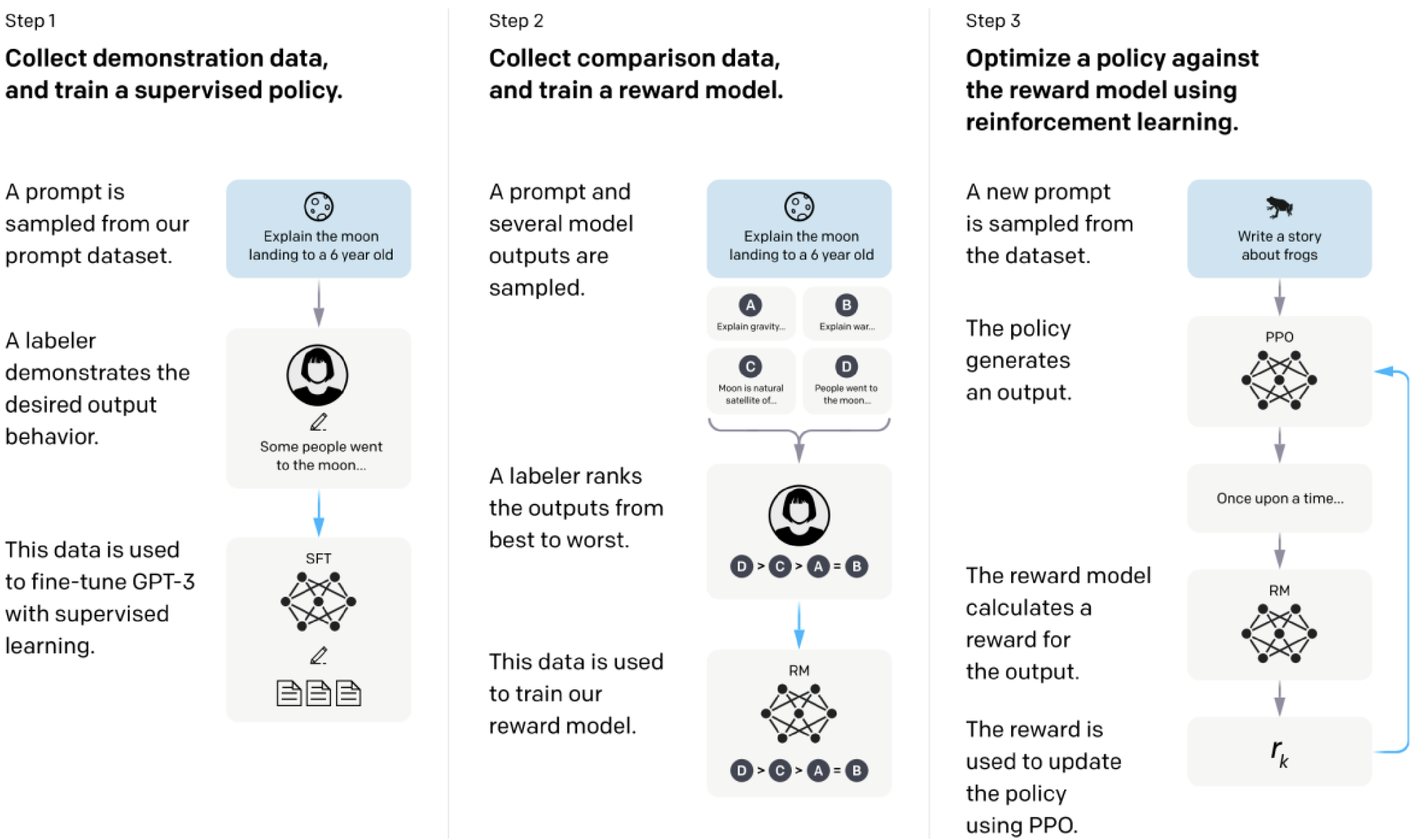
SAN FRANCISCO, California (CNN) -- A magnitude 4.2 earthquake shook the San Francisco ... overturn unstable objects.	An earthquake hit San Francisco. There was minor property damage, but no injuries.	The Bay Area has good weather but is prone to earthquakes and wildfires.
	s_1	s_2
	$R(s_1) = 8.0$	$R(s_2) = 1.2$

RLHF Pipeline



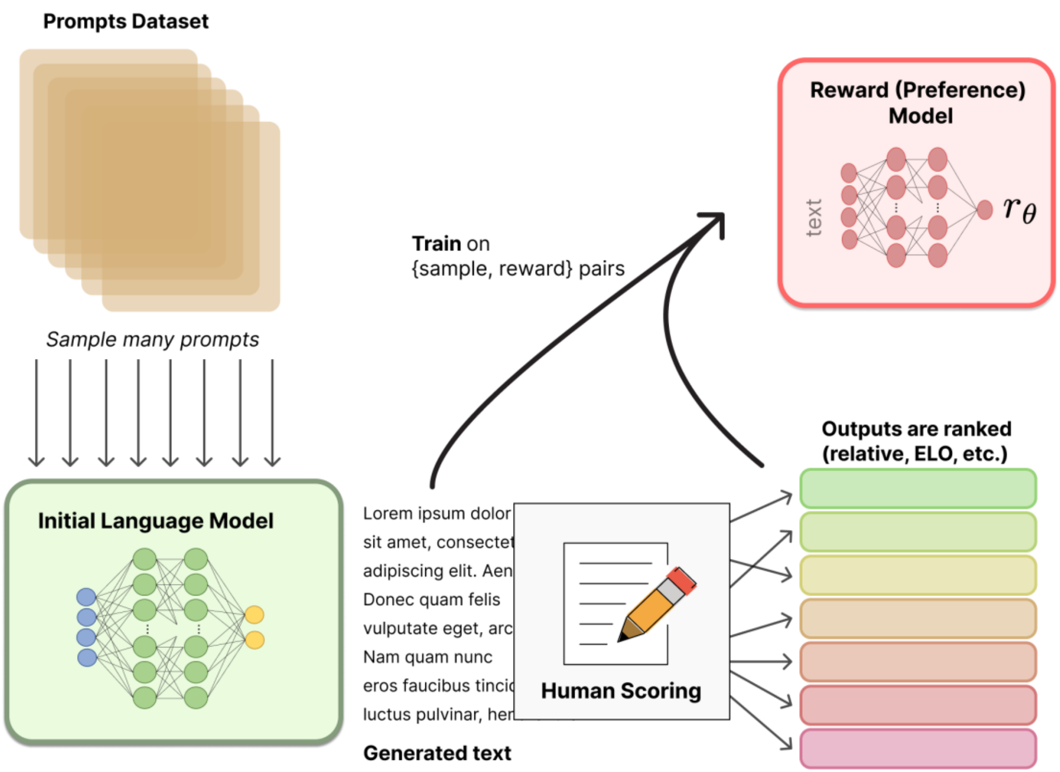
Primeiro passo: Instruction Tuning!

RLHF Pipeline



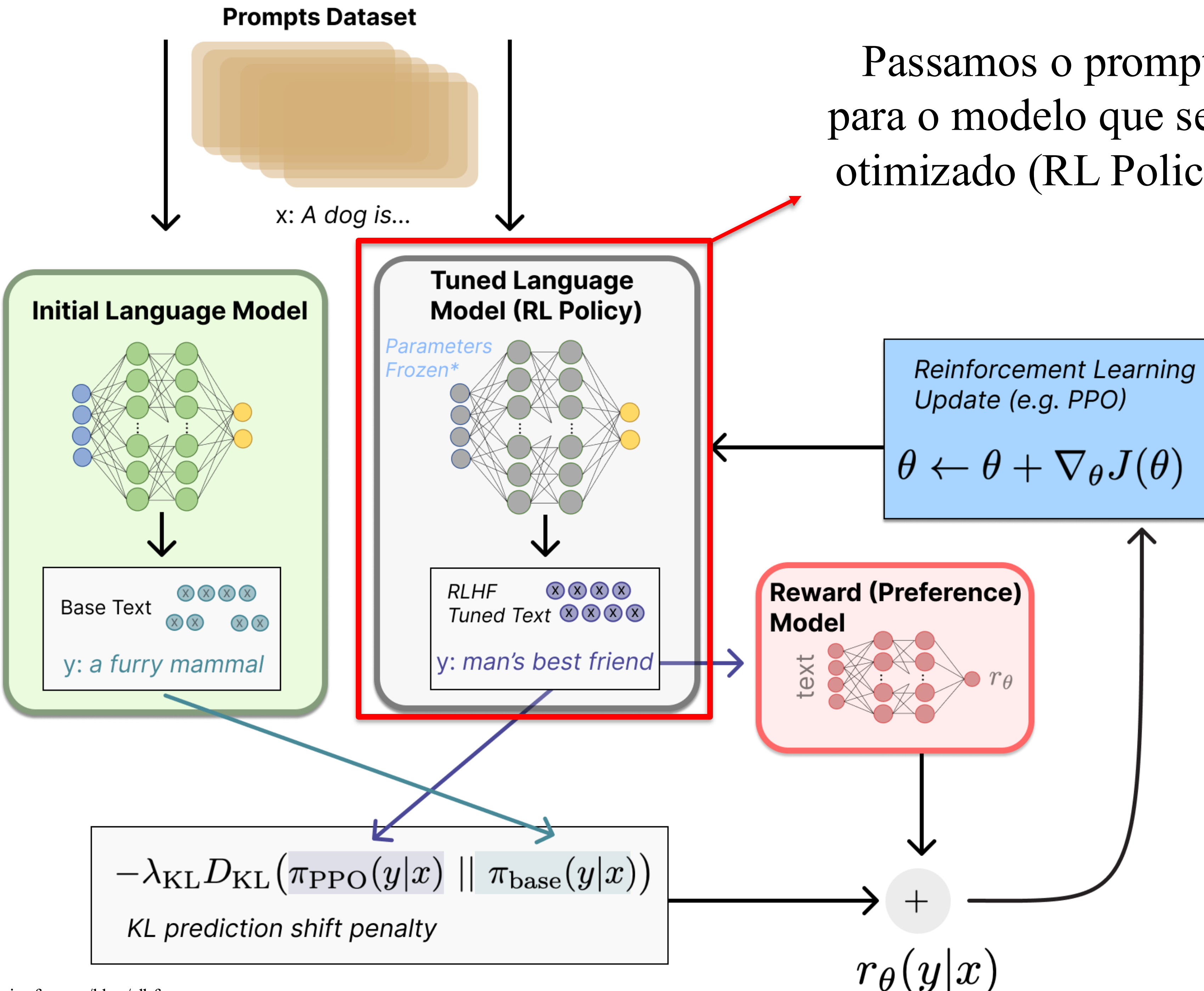
RLHF Pipeline

- Treinamos um Reward Model para apontar quais textos são preferidos (comparações binárias)



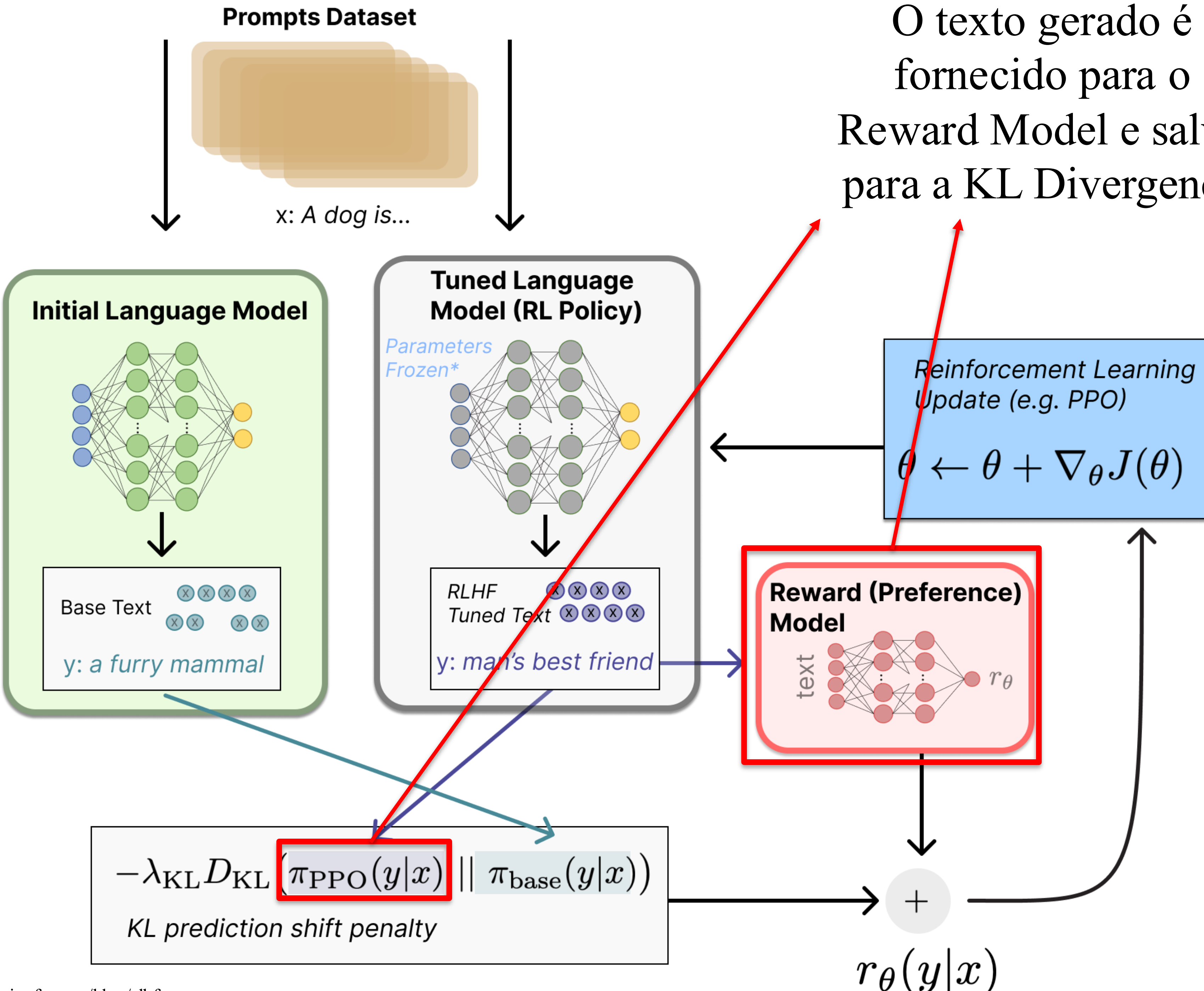
https://huggingface.co/blog/rlhf

Passamos o prompt para o modelo que será otimizado (RL Policy)



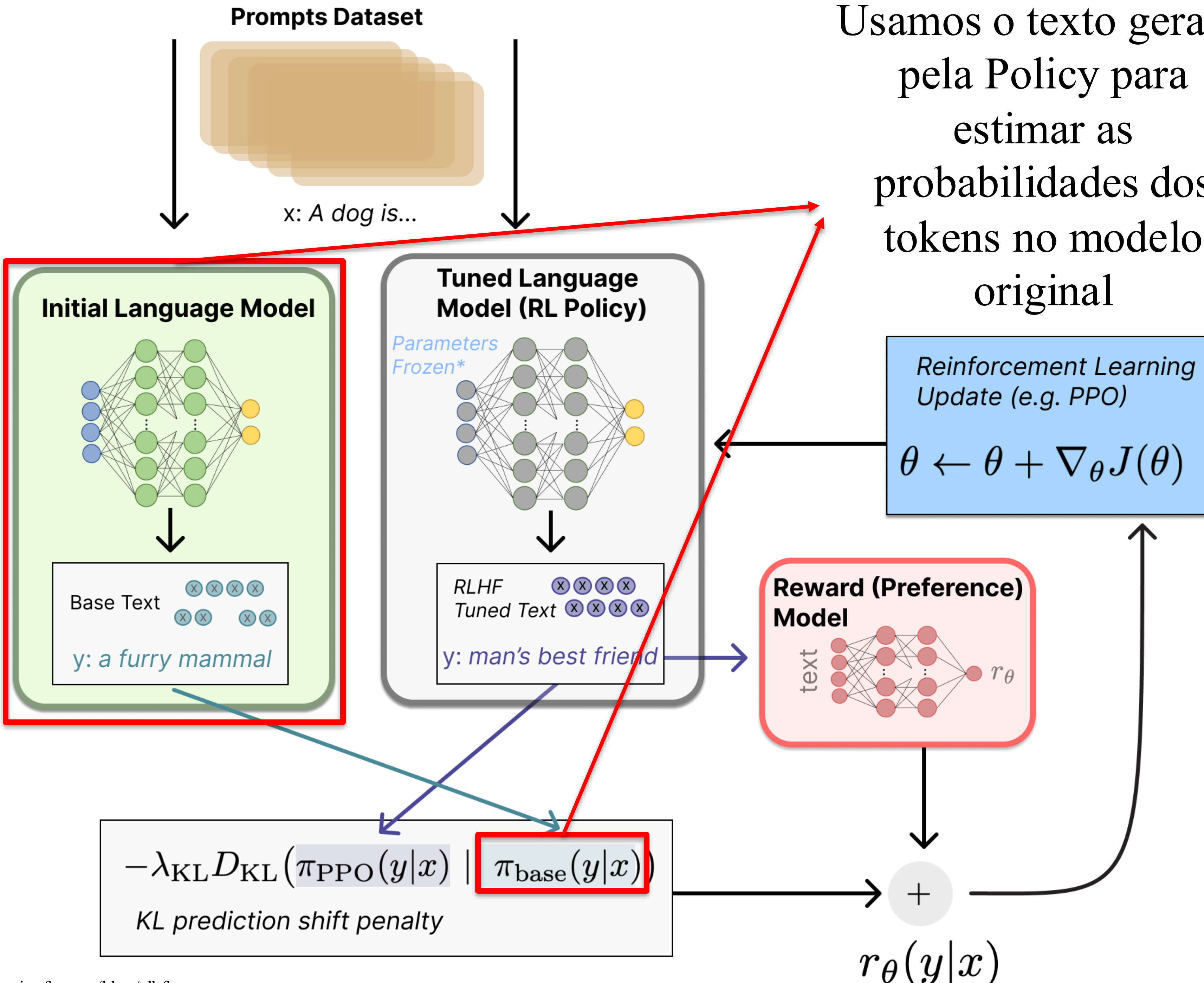
https://huggingface.co/blog/rlhf

O texto gerado é fornecido para o Reward Model e salvo para a KL Divergence



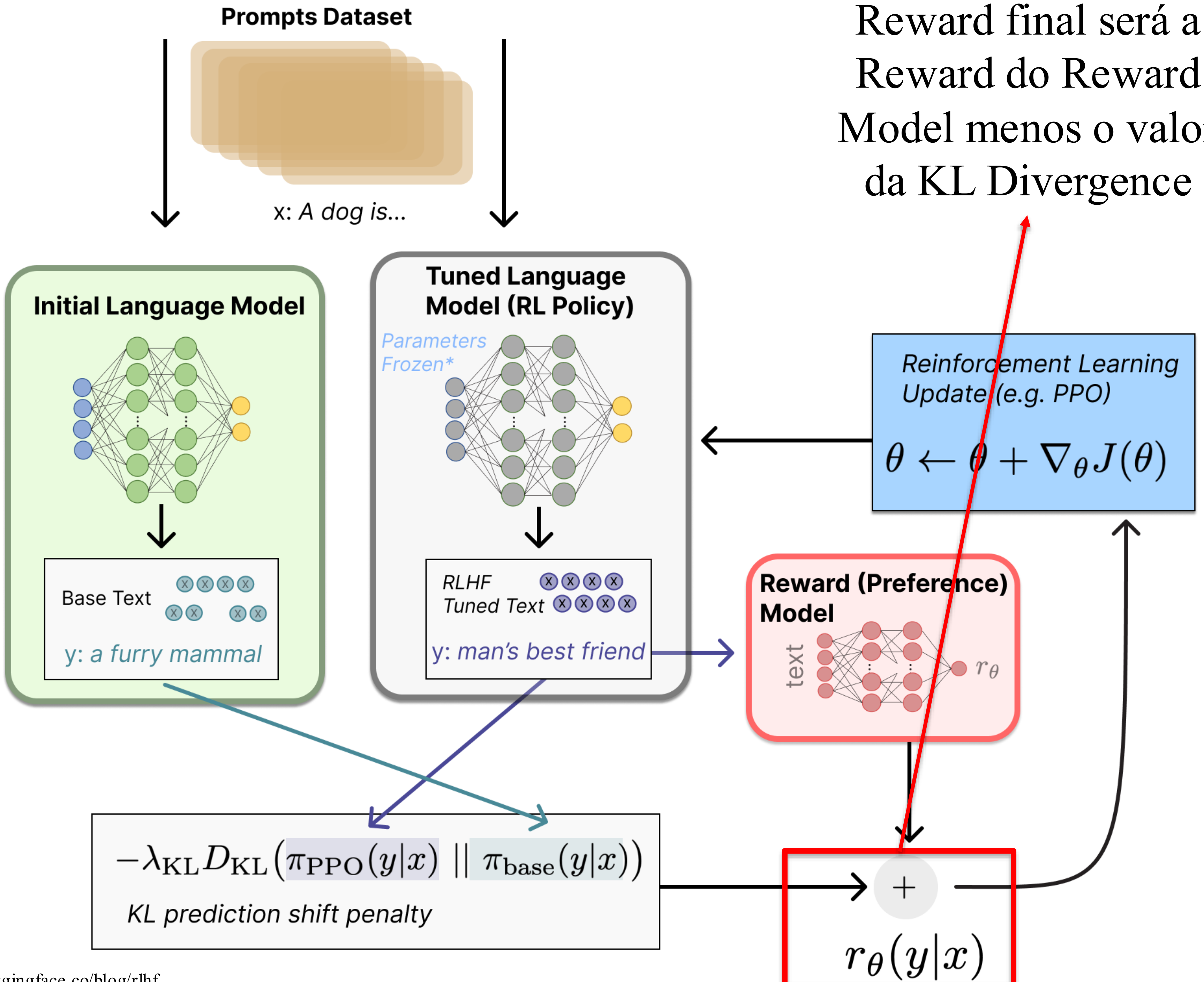
https://huggingface.co/blog/rlhf

Usamos o texto gerado pela Policy para estimar as probabilidades dos tokens no modelo original



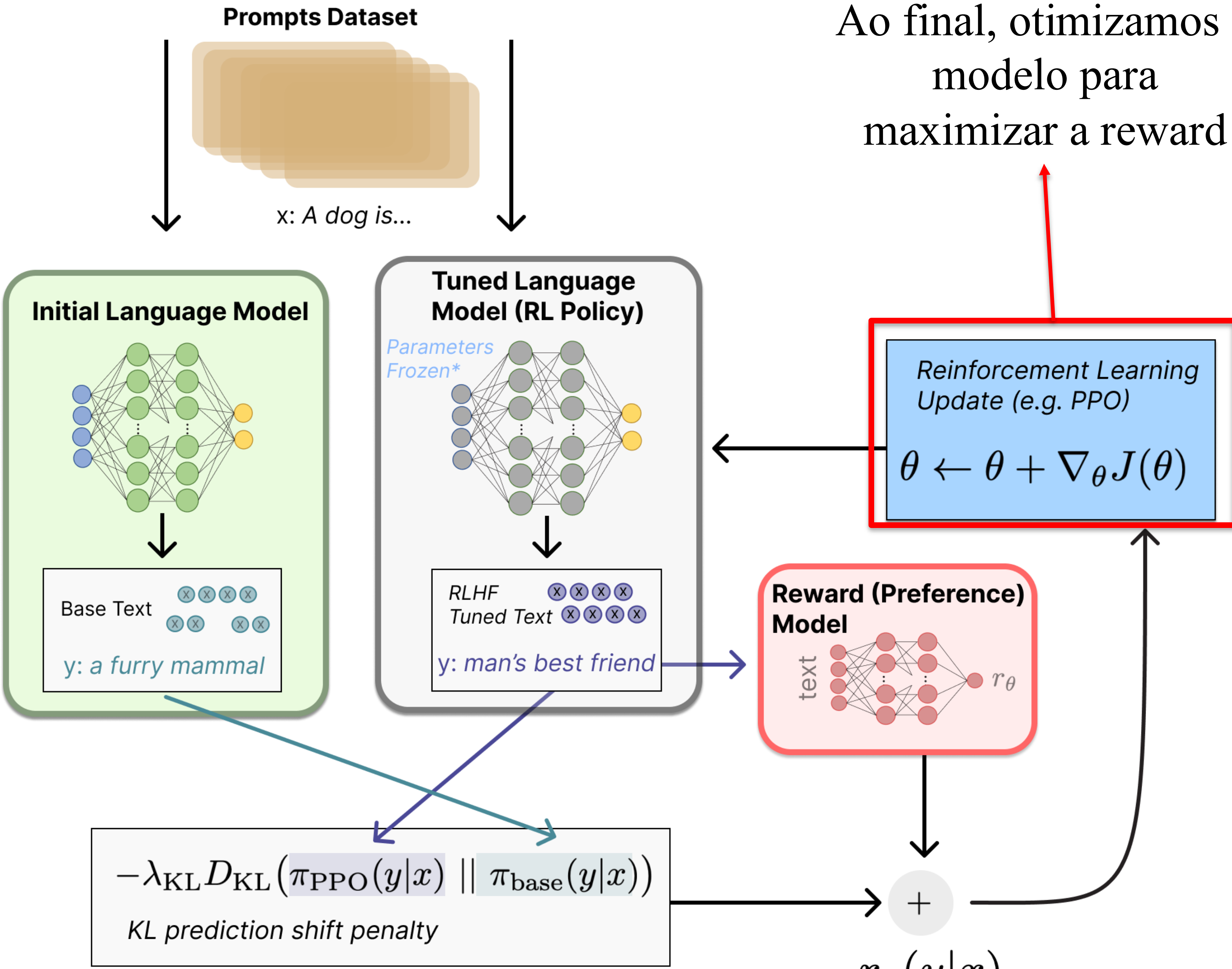
https://huggingface.co/blog/rlhf

Reward final será a Reward do Reward Model menos o valor da KL Divergence



https://huggingface.co/blog/rlhf

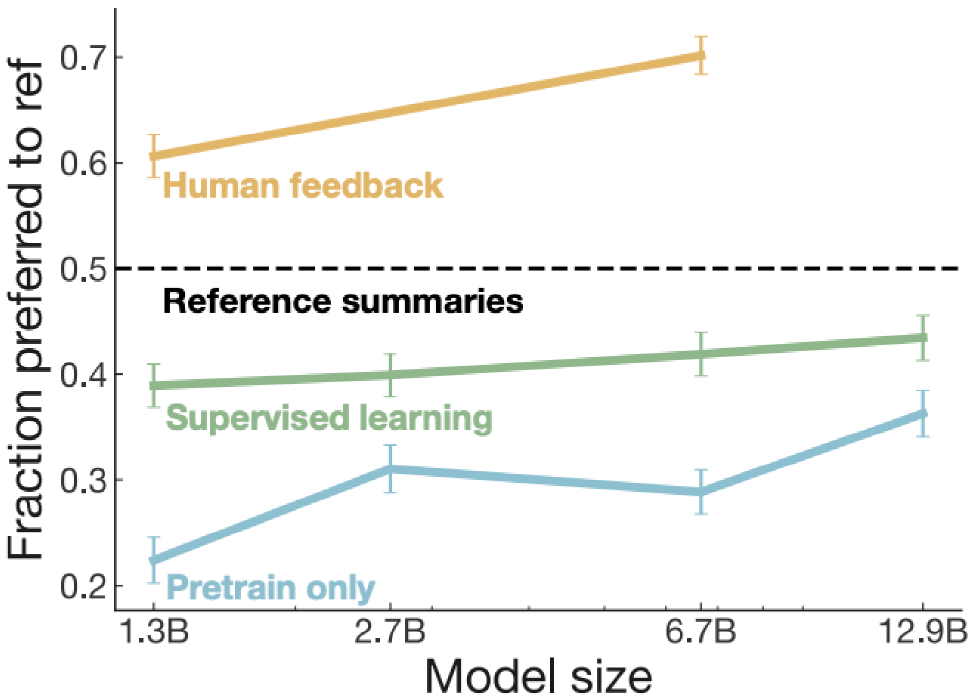
Ao final, otimizamos o modelo para maximizar a reward



https://huggingface.co/blog/rlhf

RLHF Pipeline

- RLHF foi uma ideia ótima para alinhar modelos a preferências humanas



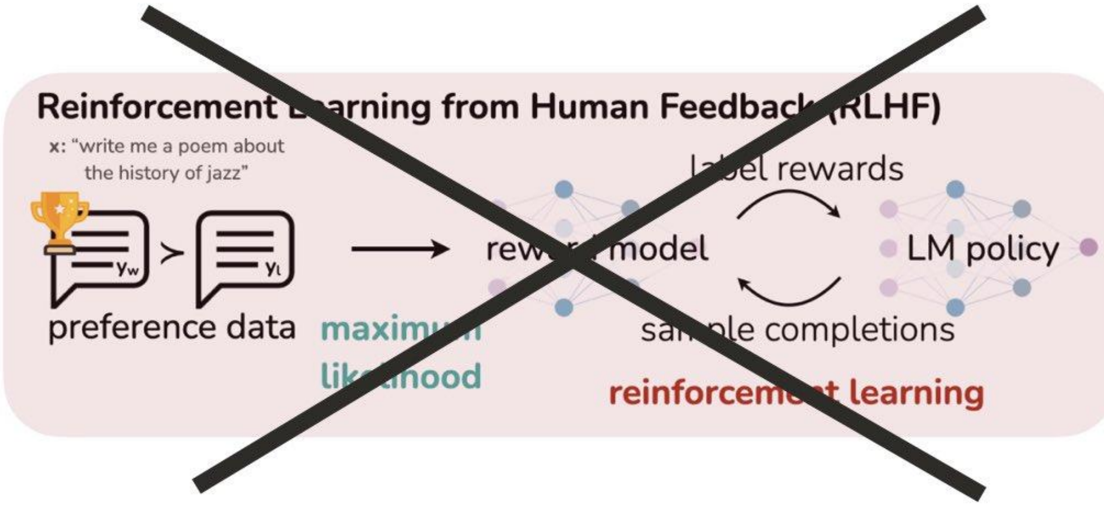
Stiennon, Nisan, et al. "Learning to summarize with human feedback." *Advances in neural information processing systems* 33 (2020): 3008-3021.

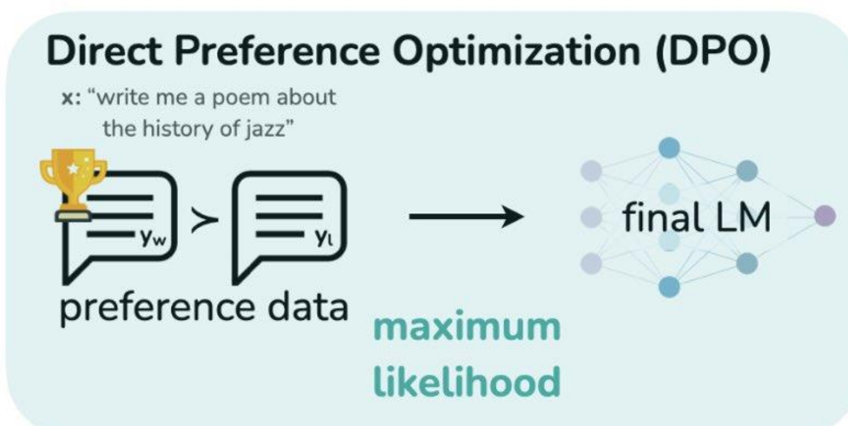
RLHF Pipeline

- Porém RLHF possui alguns problemas:
  - Realizar as anotações humanas é custoso
  - O processo é bem conturbado!
  - Pode ocorrer o que chamamos de Reward Hacking:
    - LLM aprende a maximizar a Reward, mas por caminhos não desejados

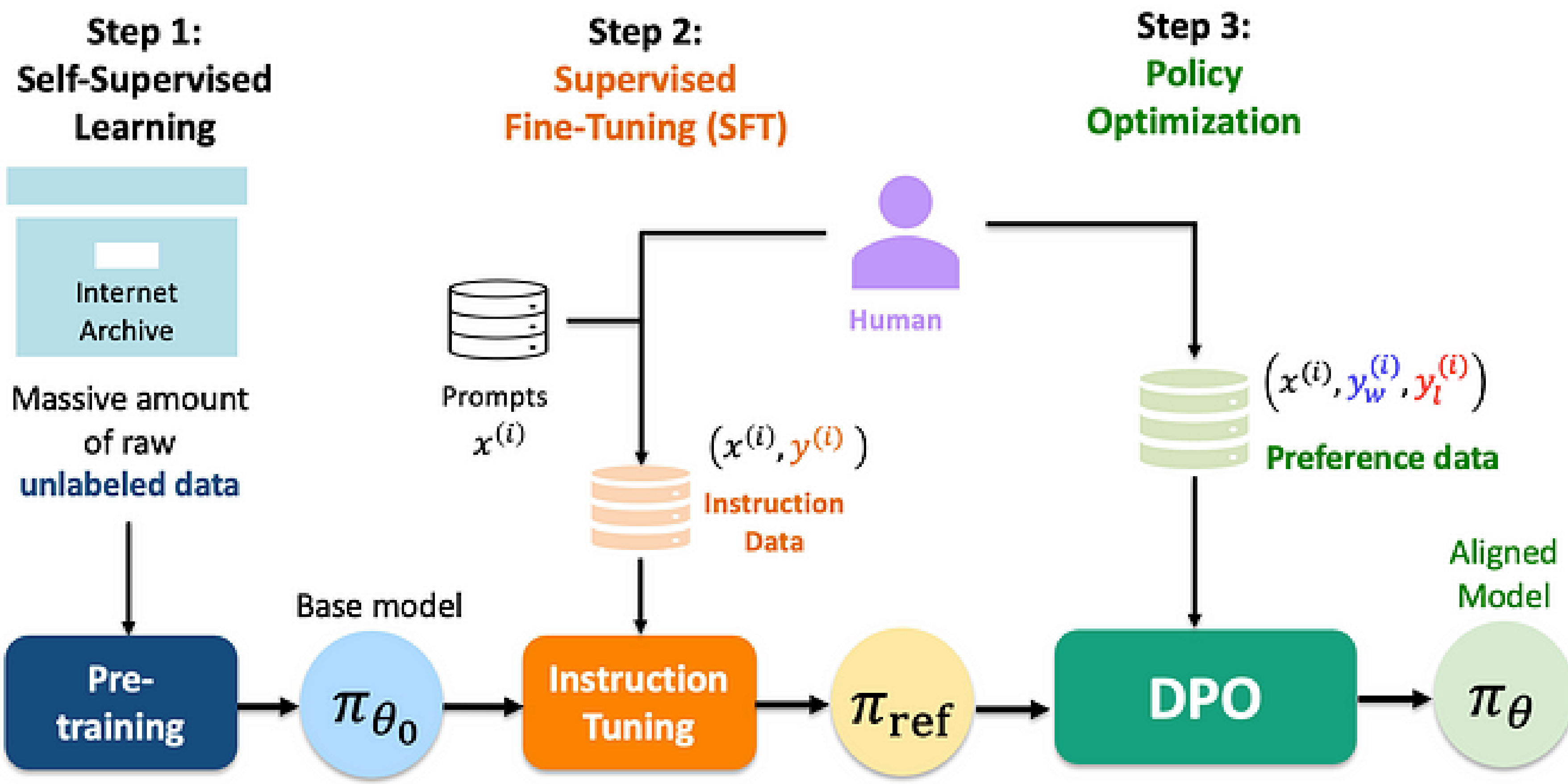
https://openai.com/index/faulty-reward-functions/

DPO


x: "write me a poem about the history of jazz"
y_w
y_l
preference data
LM policy


Direct Preference Optimization (DPO)
x: "write me a poem about the history of jazz"
y_w
y_l
preference data
maximum likelihood
final LM

DPO



https://medium.com/@lmpo/direct-preference-optimization-a-novel-approach-to-language-model-alignment-1f829d4ac306

DPO

- Na prática, não precisamos de um modelo de recompensa
- Nosso LLM já serve como um modelo de recompensa

$$\mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\text{ref}}(y_l \mid x)} \right) \right]$$

Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." *Advances in Neural Information Processing Systems* 36 (2023): 53728-53741.

DPO

- Na prática, não precisamos de um modelo de recompensa
- Nosso LLM já serve como um modelo de recompensa

$$-\boxed{\mathbb{E}_{(x,y_w,y_l)\sim\mathcal{D}}}$$

Nosso dataset consiste de exemplos bons (win) e ruins (lose)

Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." *Advances in Neural Information Processing Systems* 36 (2023): 53728-53741.

DPO

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- Nosso LLM já serve como um modelo de recompensa

$$\mathcal{L}_{\text{DPO}}\boxed{(\pi_\theta; \pi_{\text{ref}})}$$

Ainda teremos dois modelos instanciados, o original e o que será otimizado

Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." *Advances in Neural Information Processing Systems* 36 (2023): 53728-53741.

DPO

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Queremos aumentar a probabilidade do modelo otimizado a gerar os tokens da sentença positiva e diminuir a de gerar os tokens da sentença negativa

Rafailov, Rafael, et al. "Direct preference optimization: Your language model is secretly a reward model." *Advances in Neural Information Processing Systems* 36 (2023): 53728-53741.